

Global Warming Science

<https://courses.seas.harvard.edu/climate/eli/Courses/EPS101/>

Arctic sea ice

Please use the template below to answer the workshop questions. “XX” indicates places where you need to complete/write code or add a discussion.

your name:

```
[ ]: # import needed libraries:
# -----
import numpy as np
import scipy as sp
import scipy.ndimage
import matplotlib.pyplot as plt
import matplotlib
from matplotlib.colors import BoundaryNorm
import pickle
import cartopy.crs as ccrs
import cartopy.feature as cfeature
#from cartopy.feature import NaturalEarthFeature
from cartopy import config
from matplotlib.colors import BoundaryNorm
from matplotlib import animation, rc
from IPython.display import HTML

# load data from pickle file:
# -----
with open('./sea_ice_variables.pickle', 'rb') as file:
    d = pickle.load(file)
    # print information about each extracted variable:
    for key in list(d.keys()):
        print("extracting pickled variable: name=", key, "; size=", d[key].shape)
        #print("; type=", type(d[key]))
    globals().update(d)

# set defaulty font size:
font = {'size' : 11}
matplotlib.rc('font', **font)
```

Explanation of sea ice variables:

Data are for the Arctic unless Antarctica is denoted in variable name

sea ice extent time series:

sic_extent_obs_timeseries_Arctic, sic_extent_obs_timeseries_time_Arctic,

sic_extent_obs_timeseries_Antarctica, sic_extent_obs_timeseries_time_Antarctica

sea ice area time series:

sic_area_obs_timeseries_Arctic, sic_area_obs_timeseries_time_Arctic

sic_area_obs_timeseries_Antarctica, sic_area_obs_timeseries_time_Antarctica,

sea ice age as a function of lat/ lon and corresponding axes:

sic_age_obs1984, sic_age_obs2024, sic_age_obs_lat, sic_age_obs_lon

****sea ice area time series: (_Antarctica/_Arctic) ****

sic_area_obs_timeseries_Arctic, sic_area_obs_timeseries_time_Arctic, sic_area_obs_timeseries_Antarctica,
sic_area_obs_timeseries_time_Antarctica_time

Observed sea ice concentration in September as a function of year, lat, lon:

sic_concentration_obs_Sept_1979, sic_concentration_obs_Sept_YYYY (last year in obs), sic_concentration_obs_lat,
sic_concentration_obs_lon,

Projected sea ice concentration in September as a function of year, lat, lon:

sic_concentration_rcp85_sept, sic_concentration_rcp85_sept_lat, sic_concentration_rcp85_sept_lon, sic_concentration_

observed sea ice thickness as a function of lon/lat/month during two years:

sic_thickness_obs1979, sic_thickness_obs2024, sic_thickness_obs_lat, sic_thickness_obs_lon

and cell area used for calculations:

sic_thickness_obs_cell_area

Antarctica time series:

Antarctica_sic_area_obs_timeseries, Antarctica_sic_extent_obs_timeseries, Antarctica_sic_obs_times

A monthly time series of Arctic sea ice area in km² from a long model control integration (that is, with CO₂ fixed to pre-industrial values):

sic_CESM_control, sic_CESM_control_years

Comments:

- 1) concentration maps are in percent;
- 2) extent and area observation timeseries are in km²
- 3) thickness is in meters

1) observations over the past decades

1a) Time series: Plot the monthly sea ice area and extent (defined as the area with at least 15% sea ice cover) for the Arctic and for Antarctica, superimpose a time series of the values during the month of minimum sea ice for the Arctic (September) and maximum sea ice for Antarctica (also September):

```
[ ]: fig=plt.figure(figsize=(7,10),dpi=200)
start=8;

plt.subplot(4,1,1)
plt.plot(XX,XX,lw=0.5,label="monthly")
plt.plot(sic_area_obs_timeseries_time_Arctic[start:-1:12],XX,lw=2,label="september")
plt.xlabel("year")
plt.ylabel("sea ice area ($10^6$ km$^2$)")
plt.title("sea ice area")
plt.legend(loc="upper left");

plt.subplot(2,1,2)
```

```
plt.plot(XX,XX,lw=0.5,label="monthly")
plt.plot(XX,XX,lw=2,label="september")
plt.xlabel("year")
plt.ylabel("sea ice extent ($10^6$ km$^2$)")
plt.title("sea ice extent")
plt.legend(loc="upper left");

XX also for Antarctica!
XX find and superimpose the *maximum* month here

plt.tight_layout()
plt.show();
```

1b) Plot only the September Arctic sea ice area, superimpose a linear line fit. Plot only the September Antarctic sea ice area, superimpose horizontal lines at the location of the mean and ± 2 standard deviation calculated for the record until 2022.

```
[ ]: plt.figure(figsize=(6,8))

# minimum sea ice in NH:
line_fit_pars = np.polyfit(sic_area_obs_timeseries_time_Arctic[8::
    ↪12],sic_area_obs_timeseries_Arctic[8::12], 1)
linefit=line_fit_pars[0]*XX+XX
plt.subplot(2,1,1)
plt.plot(XX,XX,'-or',label="Observed")
plt.plot(XX,XX,'b',label="Linear fit")
plt.title("Arctic September sea ice area")
plt.xlabel("Year")
plt.ylabel("$10^6$ km$^2$")
plt.grid(lw=0.25)
plt.legend()

# maximum sea ice area in SH:
plt.subplot(2,1,1)
# calculate the mean of the record, not including the last year (2023):
linemean=XX
twostd=2*np.std(XX)
plt.subplot(2,1,2)
plt.plot(XX,XX,'-or',label="Observed")
plt.plot(XX,XX,'b',label="Mean")
plt.plot(XX,XX,'b',linewidth=1,linestyle='--',label="$\pm 2std$") # +2std
plt.plot(XX,XX,'b',linewidth=1,linestyle='--') # -2std
plt.title("Antarctic September sea ice area")
plt.xlabel("Year")
plt.ylabel("$10^6$ km$^2$")
plt.grid(lw=0.25)
plt.legend()

plt.tight_layout()
```

1c) Concentration: Compare the sea ice September minima of 1979 vs last year available by plotting the two maps.

```
[ ]: # preliminaries, defining configuration of subplots:
projection=ccrs.NorthPolarStereo(central_longitude=0.0)
fig,axes=plt.subplots(1,2,figsize=(7,4),subplot_kw={'projection': projection},dpi=200)

levels=np.arange(0,101,5)
cmap = plt.get_cmap('jet')
norm = BoundaryNorm(levels, ncolors=cmap.N, clip=True)

# concentration first year:
# -----
axes[0].set_extent([0, 359.999, 70, 90], crs=ccrs.PlateCarree())
axes[0].coastlines(resolution='110m')
axes[0].gridlines()
c=axes[0].pcolormesh(sic_concentration_obs_lon,sic_concentration_obs_lat\
                    , sic_concentration_obs_Sept_1979*100\
                    ,cmap=cmap,norm=norm,transform=ccrs.PlateCarree())
clb.set_label('concentration')
clb.set_ticks(np.arange(0,105,10))
axes[0].set_title('concentration obs, year=1979');

# concentration obs last year:
# -----
axes[1].set_extent([0, 359.999, 70, 90], crs=ccrs.PlateCarree())
axes[1].coastlines(resolution='110m')
axes[1].gridlines()
# XX
clb.set_label('concentration')
clb.set_ticks(np.arange(0,105,10))
axes[1].set_title('concentration obs, year=2024'));

# colorbar:
hcb=fig.colorbar(c, shrink=0.7, ax=axes.ravel().tolist())
hcb.set_label('Concentration (%)')
hcb.set_ticks(np.arange(0,105,20))

# finalize and show plot:
# plt.subplots_adjust(top=0.92, bottom=0.08, left=0.01, right=0.95, hspace=0.
#    15,wspace=-0.03)
plt.show()
```

1d) Thickness: Plot March sea ice thickness maps for 1979 and 2024. Calculate and plot the PDF (probability distribution function) of ice thickness for these two years for the entire year.

```
[ ]: # preliminaries, defining configuration of subplots:
projection=ccrs.NorthPolarStereo(central_longitude=0.0)
fig,axes=plt.subplots(1,2,figsize=(7,4),subplot_kw={'projection': projection},dpi=200)

levels=np.arange(0,5.3,0.2)
cmap = plt.get_cmap('jet')
norm = BoundaryNorm(levels, ncolors=cmap.N, clip=True)

# thickness first year:
# -----
```

```

axes[0].set_extent([0, 359.999, 70, 90], crs=ccrs.PlateCarree())
#XX
clb.set_label('thickness (m)')
axes[0].set_title('thickness 1979');

# thickness obs last year:
# -----
axes[1].set_extent([0, 359.999, 70, 90], crs=ccrs.PlateCarree())
#XX
clb.set_label('thickness (m)')
axes[1].set_title('thickness, 2024');

# finalize and show plot:
#plt.subplots_adjust(top=0.92, bottom=0.08, left=0.01, right=0.95, hspace=0.
↳15, wspace=-0.03)
plt.show()

```

```

[ ]: # ice thickness histogram (probabiligy distribution function):

def plot_thickness_histogram(thickness_array, title):
    # consider only latitudes north of 70N (use variable sic_thickness_obs_lat
    # loaded above from pickle file), and ignore thicknesses less than 0.1.
    # hint: use something like: a[b>3]=np.nan and a[a>3]=np.nan
    thickness=1.0*thickness_array
    thickness[XX]=np.nan
    thickness[XX]=np.nan
    # reshape thickness into a 1d vector:
    thickness.shape=XX # XX this should be set to the product of the dimensions of
↳thickness
    # calculate and plot the histogram:
    hist, bin_edges = np.histogram(thickness[~np.isnan(thickness)], bins=50,
↳density=True)
    x=(bin_edges[1:]+bin_edges[0:-1])/2
    # bar plot:
    width=0.9*(bin_edges[2]-bin_edges[1])
    plt.bar(x, hist, width=width);
    plt.xlabel("thickness (m)")
    plt.ylabel("probability")
    plt.title(title)

plt.figure(figsize=(8,4), dpi=200)
plt.subplot(1,2,1)
plot_thickness_histogram(XX, "thickness histogram 1979")
plt.subplot(1,2,2)
plot_thickness_histogram(XX, "thickness histogram 2024")
plt.tight_layout()

```

1e) Age: Plot sea ice age maps for 1984 and 2024. Calculate and plot the PDF (probability distribution function) of ice age for these two years.

```

[ ]: # preliminaries, defining configuration of subplots:
projection=ccrs.NorthPolarStereographic(central_longitude=0.0)
fig, axes=plt.subplots(1,2, figsize=(7,4), subplot_kw={'projection': projection}, dpi=200)

```

```

levels=np.arange(0,7.1,0.2)
cmap = plt.get_cmap('jet')
norm = BoundaryNorm(levels, ncolors=cmap.N, clip=True)

# thickness first year:
# -----
axes[0].set_extent([0, 359.999, 70, 90], crs=ccrs.PlateCarree())
#XX clb.set_label('age (years)')
axes[0].set_title('age, 1984');

# thickness obs last year:
# -----
axes[1].set_extent([0, 359.999, 70, 90], crs=ccrs.PlateCarree())
#XX
clb.set_label('age (years)')
axes[1].set_title('age 2024');

plt.show();

```

```

[ ]: # ice age histogram (probabiligy distribution function):

def plot_age_histogram(age_array,title):
    # consider only latitude north of 70N (use variable sic_age_obs_lat
    # loaded from pickle file), and ages larger than 0.1, smaller than 15
    age=1.0*age_array
    age[XX]=np.nan
    age[XX]=np.nan
    # reshape age into a vector:
    age.shape=XX # XX this should be set to the product of the dimensions of age
    # calculate and plot the histogram:
    hist, bin_edges = np.histogram(age[~np.isnan(age)], bins=50, density=True)
    x=(bin_edges[1:]+bin_edges[0:-1])/2
    # bar plot:
    plt.bar(x,hist);
    plt.xlim(0,10)
    plt.xlabel("age (years)")
    plt.ylabel("probability")
    plt.title(title)

plt.figure(figsize=(8,4),dpi=300)
plt.subplot(1,2,1)
plot_age_histogram(XX,"age histogram 1984")
plt.subplot(1,2,2)
plot_age_histogram(XX,"age histogram 2024")
plt.tight_layout()

```

2) How much heat is needed to melt sea ice:

Calculate how much heat per unit area (Watts/m²) is needed during the three summer months every year to completely melt a sea ice cover within 30 yr. Assume that the initial ice thickness is 3 m, that it has an initial average temperature of -3°C , and that the ocean mixed layer (the upper 50 m) is initially at the freezing temperature (-1.8°C). At the end of the period, the ocean mixed layer is at a final temperature of $T_{final} = 1.2^{\circ}\text{C}$ and the ice is completely melted. Use the following guidelines:

(a) Calculate separately the heat flux required for (i) warming the ice to its melting temperature of 0°C , (ii) melting the ice, (iii) warming the melt water to T_{final} , and (iv) warming the mixed layer to T_{final} .

(b) Find out the Arctic summer downward SW, and calculate what percent change in cloud or ice albedo during the peak summer months could account for a change in absorbed solar flux that can lead to the above complete ice melting in 30 yr. Discuss the implications for identifying the cause of Arctic sea ice melting.

```
[ ]: # some needed constants:
year=365*24*3600
L_melt=XX # latent heat of melting of ice, J/kg
Cp_ice=XX # specific heat of ice J/(kg*K)
Cp_water=XX # specific heat of water J/(kg*K)
thickness_ice=XX
thickness_water=XX
rho_ice=900
rho_water=1025

# parameter of the problem:
time=30*year*(3/12) # three months a year, for 30 years
deltaT_ocean=XX
T_final=XX
deltaT_ice=XX # warm the ice from its initial temperature to the melting temperature
meltwater_mass=XX

# 2a) calculation of the results:
flux_for_warming_ice_to_melting=XX
flux_for_melting_ice=XX
flux_for_warming_mixed_layer_water=XX
flux_for_warming_meltwater=XX

print(" flux_for_warming_ice_to_melting=%3.3g W/m^2" \
      % (flux_for_warming_ice_to_melting))
print(" flux_for_melting_ice=%3.3g W/m^2" \
      % (flux_for_melting_ice))
print(" flux_for_warming_mixed_layer_water=%3.3g W/m^2" \
      % (flux_for_warming_mixed_layer_water))
print(" flux_for_warming_meltwater=%3.3g W/m^2" \
      % (flux_for_warming_meltwater))

# 2b) what ice albedo change would cause this:
SW_radiation_summer_arctic=400 # W/m^2, see http://www.climate.be/textbook/
    ↪chapter2_node5_3.xml figure 2.11
# so less than %1 albedo change can melt the ice
print(" percent change for melting sea ice = %3.3g" \
      % XX);
```

3) Thickness-insulation feedback:

solve the equation

$$\rho_{ice} L_f \frac{dh}{dt} = \kappa \frac{T_f - T_{surface}}{h}$$

analytically for the ice thickness $h(t)$, using $T_f = -1.8^{\circ}\text{C}$ and $T_{surface} = -15^{\circ}\text{C}$, for the initial conditions $h(t_0) = h_0 = 0.1$ m. Plot the solution for one year, use it to find the time to grow the first meter of thickness

and then the second, and explain the results in terms of the negative thickness-insulation feedback.

Analytical solution:

XX

```
[ ]: # plotting:  
XX
```

4) Increasing roughness with age, melt ponds and albedo:

- (i) Explain why ice surface roughness is expected to increase with ice age.
- (ii) Suppose the surface structure as function of location x, y is given by $h(x, y) = h_0 \cos(kx)$ with $k = 2\pi/100 \text{ m}^{-1}$. Plot the area fraction as function of the melt water volume for $h_0 = 0.5 \text{ m}$ and for $h_0 = 1 \text{ m}$.
- (iii) based on your results, explain the positive feedback leading to more melting due to the decrease in ice age and therefore decrease in ice roughness.

Solution: Given that the sea ice surface is given by $h(x, y) = h_0 \cos(kx)$ with $k = 2\pi/100 \text{ m}^{-1}$, the lowest point is at $kx = \pi$. Consider a single wavelength and suppose the melt water fill the low point from $kx = (\pi - \theta)$ to $kx = (\pi + \theta)$. We can calculate the water area and volume as function of θ . Then, we can plot the area fraction as function of volume, as requested.

$$\text{melt water area fraction}(\theta) = 2\theta/(2\pi)$$

and

$$\text{melt water volume}(\theta) = - \int_{(\pi-\theta)/k}^{(\pi+\theta)/k} h_0 \cos(kx') dx' = XX.$$

```
[ ]: # calculate and plot the above solution:  
k=2*np.pi/100  
h1=0.5 # m, amplitude of ice surface height variations  
h2=1 # m, amplitude of ice surface height variations  
theta=np.arange(0,np.pi/2,0.01)  
area_fraction=theta/np.pi  
volume1=XX# vector of melt water volume calculated for h1  
volume2=XX # vector of melt water volume calculated for h2  
  
# first, plot everything as function of theta:  
# -----  
fig = plt.figure(dpi=200)  
ax1 = fig.add_subplot(111)  
ax1.plot(XX,XX,color="r",label="area fraction")  
ax1.set_xlabel("$\\theta$ (degrees)")  
ax1.set_ylabel("melt pond area fraction",color="r")  
  
ax2 = ax1.twinx() # instantiate a second axes that shares the same x-axis  
ax2.plot(theta*180/np.pi,volume1,color="b",label="volume h=0.5m");  
ax2.plot(theta*180/np.pi,volume2,linestyle="--",color="b",label="volume h=2m");  
ax2.set_ylabel("volume",color="b")  
  
fig.legend(loc="upper left");  
ax1.grid('both')
```

```
plt.tight_layout()

# now, plot area fraction covered by melt water as function of melt water volume:
# -----
plt.figure(dpi=150)
plt.plot(XX,XX,label="volume h=0.5m")
plt.plot(XX,XX,label="volume h=2m")
plt.xlabel("melt water volume")
plt.ylabel("melt pond area fraction")
plt.title("melt pond area fraction (1-albedo) as func of melt water volume")
plt.legend();
```

XX discussion

5) Projections: Contour the September sea ice concentration at the beginning of the 21st century and at year 2100 according to the RCP8.5 scenario.

```
[ ]: # preliminaries, defining configuration of subplots:
projection=ccrs.NorthPolarStereo(central_longitude=0.0)
fig,axes=plt.subplots(1,2,figsize=(12,6),subplot_kw={'projection': projection},dpi=200)

levels=np.arange(0,101,1)
cmap = plt.get_cmap('jet')
norm = BoundaryNorm(levels, ncolors=cmap.N, clip=True)

# thickness first year:
# -----
axes[0].set_extent([0, 359.999, 70, 90], crs=ccrs.PlateCarree())
#XX
clb.set_label('concentration')
clb.set_ticks(np.arange(0,105,10))
axes[0].set_title('sic_concentration_rcp85_sept_
↳'+repr(sic_concentration_rcp85_sept_years[0]));

# thickness obs last year:
# -----
axes[1].set_extent([0, 359.999, 70, 90], crs=ccrs.PlateCarree())
#XX
clb.set_label('concentration')
clb.set_ticks(np.arange(0,105,10))

axes[1].set_title('sic_concentration_rcp85_sept_
↳'+repr(sic_concentration_rcp85_sept_years[-1]));

# finalize and show plot:
#plt.subplots_adjust(top=0.92, bottom=0.08, left=0.01, right=0.95, hspace=0.
↳15,wspace=-0.03)
plt.show()
```

6) Climate change detection:

(a) Examine if the trend in annual minimum sea ice area over the satellite period since 1979 is consistent with natural variability, as estimated using a long run of a climate model with no anthropogenic greenhouse forcing. Calculate the PDF for a period corresponding to the length of the observed record, and calculate the probability that the observed trend is part of natural variability. Repeat the calculation using only the record from 1979 to 1999; what differences do you see between the trends during the two periods and the probability that they may be due to natural variability alone?

```
[ ]: # calculate the observed trend over the past 40 years,
# normalized to millions of km2 per 10 years:
# -----
start=8
sic_area_obs_timeseries_sept_time_Arctic=sic_area_obs_timeseries_time_Arctic[XX:XX:XX]
sic_area_obs_timeseries_sept_Arctic=sic_area_obs_timeseries_Arctic[XX:XX:XX]
interval_obs=XX # number of years in observed satellite record
Nyears_obs=int(interval_obs)
x=1.0*sic_area_obs_timeseries_sept_time_Arctic
y=1.0*sic_area_obs_timeseries_sept_Arctic
fit_obs = np.polyfit(x, y, 1)
fit_obs_line=(x*fit_obs[0]+fit_obs[1])
trend_obs=10*fit_obs[0]
print("observed decadal trend = ",trend_obs,"over a period of" \
      ,interval_obs," years.")

# plot fit to observed sept sea ice:
fig=plt.figure(figsize=(4,3),dpi=300)
plt.plot(XX,XX,label="Observed September sea ice")
plt.plot(XX,XX,label="Linear fit")
plt.xlabel("Year")
plt.ylabel("Sea ice area ($10^6$ km$^2$)")
plt.legend()
plt.tight_layout()
plt.show()
```

```
[ ]: # Calculate the PDF of trends from the control model run for intervals
# of the length of the observed period:
# -----

# get september values only:
N_months=len(sic_CESM_control)
start_month=8
sic_CESM_control_sept=sic_CESM_control[XX:XX:XX]*1.e6
sic_CESM_control_sept_years=sic_CESM_control_years[XX:XX:XX]
Nyears=len(sic_CESM_control_sept)

# define bins for trend histogram:
Dtrend=0.05 # resolution of bins of trend amplitudes
bin_edges=np.arange(-1,1.00001+Dtrend,Dtrend)
# loop over time intervals from 2 to 100 years
interval=40
# loop over all start years for intervals, calculate and accumulate trends:
trends=[]
```

```

for start_year in range(0,Nyears-interval-1):
    # for each start year, calculate the trend at the
    # interval being considered using a line fit, normalized to
    # trend of  $10^6$  km2 per 10 years:
    x1=np.arange(start_year,(start_year+interval),1.0)
    y1=XX # sea ice area over same years
    line_fit_pars = np.polyfit(x1, y1, 1) # fit a line, specify polynomial degree
    trends.append(line_fit_pars[0]*10/1.e6) # convert to units of million km per decade

# bin and plot trends
hist,bin_edges_output=np.histogram(XX,XX,density=True)
PDF_at_observed_period=hist

# verify that PDF sums to 1:
print("integrated PDF at observed interval=",np.sum(XX))

# calculate the probability that the observed trend is due to
# natural variability:
probability_of_obs_trend=sum(PDF_at_observed_period[bin_edges[0:-1]<XX]*Dtrend)
print("probability that observed trend over %d years is due to natural variability=%g"%
    \
    (interval,probability_of_obs_trend))

```

```

[ ]: # A line plot of the pdf for interval=number of observed years:
bin_centers=0.5*(bin_edges[0:-1]+bin_edges[1:])
fig=plt.figure(figsize=(5,3),dpi=300)
plt.plot(XX,XX)

# add a vertical bar at the location of the observed trend:
plt.plot(XX,XX,linewidth=3,color="grey")

plt.xlabel("Decadal trend ( $10^6$  km2/10 years)")
plt.ylabel("PDF")

plt.tight_layout()
plt.show()
fig.savefig("Output/sea-ice-area-trends-PDF-CESM-control-run-for-40-year-intervals.
pdf")

```

XX discussion

optional:

```

[ ]: # calculate the PDF as a function of both interval and trend:
# -----
# define bins for trend histogram:
bin_edges=np.arange(-1,1.06,0.05)*1
trend_intervals=np.arange(2,101,1)
# initialize trends probability distribution function as function of interval and
# trend:
trends_to_contour=np.zeros((len(trend_intervals),len(bin_edges)-1))
# loop over time intervals from 1 to 100 years
i_interval=-1
for interval in trend_intervals:

```

```

i_interval=i_interval+1
# loop over all start years for intervals, calculate and accumulate trends:
trends=[]
for start_year in range(0,Nyears-interval-1):
    # for each start year, calculate the trend at the
    # interval being considered using a line fit, normalized to
    # trend of 106 km2 per 10 years:
    x1=np.arange(start_year,(start_year+interval),1.0)
    y1=XX
    line_fit_pars = np.polyfit(x1, y1, XX) # fit a line, specify polynomial degree
    trends.append(line_fit_pars[0]*XX) # convert to units of million km per decade
# bin trends
#print(interval,"trends=",trends)
hist,bin_edges_output=np.histogram(trends,bin_edges)
# save bins to be plotted, normalized to fraction of total number
# of bins for the given interval
max_bins=Nyears-interval-1
trends_to_contour[i_interval,:]=hist/max_bins

# contour the calculated pdf for trends:
# note that in the solutions manual, the PDF is gaussian-smoothed and therefore looks
↳ a bit different.
trend_amplitude=bin_edges # units of million km2/10 years
fig=plt.figure(figsize=(5,4),dpi=300)
levels=np.log10(np.array([0.001,0.00316,0.01,0.0316,0.1,0.316,1,3.16,10]))
plt.contourf(trend_intervals,bin_edges[0:-1],np.log10(trends_to_contour.T),levels)
plt.ylim(-1,0)
plt.title("log10(pdf) of trends")
plt.xlabel("trend interval (years)")
plt.ylabel("trend (106 km2/10 years)");
plt.colorbar();

# now plot a large x at the location of the observed trend over the
# satellite period, normalized to millions of km2 per 10 years:
interval_obs=(sic_area_obs_timeseries_time_Arctic[-1]-sic_area_obs_timeseries_time_Arctic[0])
trend_obs=XX
plt.plot(interval_obs,trend_obs,'x',markersize=16,color='r',label="observed trend");

```

7) Projections.

Contour the September sea ice concentration maps in the RCP8.5 projection every five years. How old will you be when September sea ice vanishes according to this projection?

```

[ ]: fig, axes = plt.subplots(nrows=4, ncols=5, figsize=(15, 9), constrained_layout=True,
                             subplot_kw={'projection': ccrs.
↳ NorthPolarStereo(central_longitude=0)})
ax=axes.flat

for i, t in enumerate(sic_concentration_rcp85_sept_years):
    # Contour:
    # -----
    ax[i].set_extent([0, 359.999, 55, 90], crs=ccrs.PlateCarree())

```

```
ax[i].coastlines(resolution='110m')
ax[i].gridlines()

cmap = plt.cm.jet.copy()
cmap.set_bad(color='white') # make NaNs white

c=ax[i].pcolormesh(XX, XX, XX \ # lon, lat, data
                  ,transform=ccrs.PlateCarree(), cmap='jet')
ax[i].set_title('Year = '+XX)
cb = plt.colorbar(c, ax=ax[i], shrink=0.85, pad=0.02)

fig.suptitle("XX", fontsize=16);
```

[]: